

Economic History II

Bureaucracy and public service delivery

FGV EPGE

- **Opening** (classes 1, 2)
 - Reversal of fortune in the colonies
 - Colonial origins and causality
- **Institutions** (3)
 - Trade and tolerance in medieval India
- **Colonialism** (4 to 7)
 - The slave trades and Africa
 - The slaughter of the bison
 - Colonial medicine in Central Africa
 - Jesuit missions in South America
- **Antiquity** (8 to 10)
 - Seasonality and the origin of agriculture
 - Assyrian merchants at Kanesh
- **Leadership** (11, 12)
 - Monarchs and state performance
 - Heroic networks in Vichy France
- **Slavery in Brazil**, guest classes (13, 14)
 - The economics of slavery
 - The political economy of abolition
- **Bureaucracy** (16 to 18)
 - Patronage in the British Empire
 - The civil service exam in China
 - US civil service reform and Embrapa (today)
- **Reformation** (19, 20)
 - Luther and Suleyman
 - The dissolution of the monasteries
- **Two revolutions**, guest classes (21, 22)
 - Industrial Revolution and useful knowledge
 - French Revolution and extractive taxation
- **Political identity** (23 to 25)
 - Schools and the making of France
 - Abolitionism
 - Cross-border radio and nationalism
- **Student presentations** (26 to 29)

Outline of the class

- **Three papers**, taken separately. All on bureaucrats and what the state delivers.
- Aneja and Xu (2024): civil service reform in the US post office during the Gilded Age. *American Economic Review*.
- Moreira and Pérez (2024): the same reform in the customs houses, briefly. *AEJ: Applied*.
- A detour to the present: judges and school principals in Brazil, hiring in firms.
- Akerman, Moscona, Pellegrina and Sastry (2025): Embrapa and Brazil's agricultural revolution. Working paper.

Post offices, patronage and reform

Aneja and Xu (2024), *Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age*, AER

The question

- An impartial civil service with lifelong careers: the standard prescription for a working state, since Weber.
- Direct evidence that such reforms improve state effectiveness: scarce.
- The setting is the **Pendleton Act of 1883**, the first nationwide civil service reform in the United States.
- The organization is the post office, the largest civilian agency of the federal government.

Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age[†]

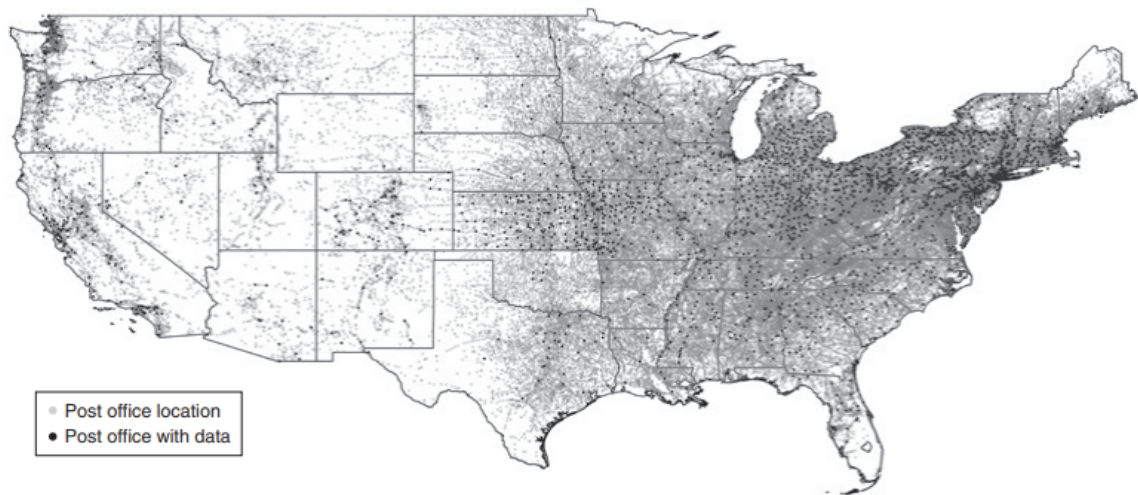
By ABHAY ANEJA AND GUO XU*

We use newly digitized records from the post office to study the effects of strengthened state capacity between 1875 and 1901. Exploiting the implementation of the Pendleton Act—a landmark statute that shielded bureaucrats from political interference—across US cities over two waves, we find that civil service reform reduced postal delivery errors and increased productivity. These improvements were most pronounced during election years when the reform dampened bureaucratic turnover. We provide suggestive evidence that reformed cities witnessed declining local partisan newspapers. Separating politics from administration, therefore, not only improved state effectiveness but also weakened the role of local politics. (JEL D24, D73, H83, J45, L82, L87, N41)

The post office in the nineteenth century

- At the onset of the Pendleton Act, postal workers were **34 percent of the entire federal workforce**.
- Principal method of long-distance communication in an expanding nation.
- Offices in virtually all parts of the country, all performing the same function: collection and delivery of mail, by clerks and carriers.
- That uniformity is what makes performance comparable across cities and years.
- John (1995): for most Americans, “the postal system was the central government”.

Panel A. All post offices



The spoils system

- Before 1883: patronage-based hiring and firing, with elected politicians holding discretion over appointments.
- Patronage employees campaigned for their benefactors and paid part of their salary back as **political assessments**.
- Eaton (1885) on what the parties extracted: “the plunder of the salaries of the public servants for filling the party treasury”.
- Careers tied to particular politicians, so turnover was high.
- One report on the New York City Post Office: “incompetency, neglect, confusion and drunkenness”.

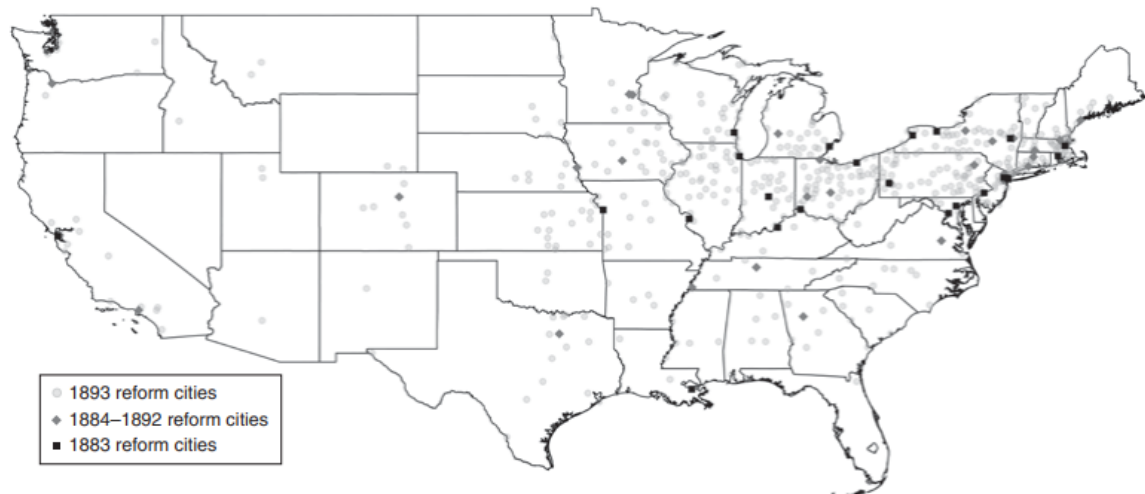
The Pendleton Act (1883)

- Discretionary appointments replaced by rule-based hiring through open, competitive **examinations**.
- Classified workers protected from removal without cause.
- Assessments outlawed for the whole federal government.
- Catalyzed by the assassination of President Garfield in 1881 by a disappointed office seeker.
- About 10 percent of federal employees were classified at the outset.

The reform arrived in waves

- Coverage was decided in Washington and keyed to size, not to local preference.
- 1883: post offices with at least 50 employees. **23 cities.**
- 1893: Harrison's executive order classified every post office with free delivery service.
556 more cities.
- In between, 30 cities crossed the 50-employee threshold on their own.
- By 1921, 80 percent of the federal civilian workforce was covered.

Panel B. Expansion of civil service reform



The data

- Large-scale digitization of the post office's own annual reports.
- **Delivery errors**: errors on returned slips, city by year, 1879 to 1901, 2,319 cities.
- Free delivery statistics: volume per carrier and cost per volume, 512 large cities, 1875 to 1891.
- Official Registers: every clerk and carrier, 1877 to 1901, 297,932 individual-year observations, linked to the full-count censuses of 1880 and 1900 (36 percent match).
- Newspaper directories: partisan and independent papers by city and election year.

C Census linking

We match the personnel records from the Official Registers of the United States (the “Registers”) to the U.S. full count Decennial Census to obtain additional individual background characteristics. Each postal worker is matched based on the full name (first name, middle name/initial, last name), birth state, and current state of employment. We proceed by using different combinations, successively relaxing the matching restrictions:

- Step 1: First name + middle name + last name + birth state + current state
- Step 2: First name + middle name initial + last name + birth state + current state
- Step 3: First name + last name + birth state + current state
- Step 4: First name + last name + birth state

Given the limited number of variables that are available for matching, we pursue a conservative approach to ensure we do not overmatch by linking incorrect individuals (i.e., false positives). To start with, we always match individual names exactly. Second, we discard candidate matches of Census respondents who were younger than

Table C1: Characteristics of Census-linked vs. non-linked workers

	(1)	(2)	(3)	(4)
	<u>Mean characteristics</u>		<u>Differences</u>	
	Matched	Unmatched	Raw	Conditional
<i>Panel A: Individual characteristics</i>				
Log(Salary)	5.80	5.83	-0.025 (0.007)	0.016 (0.007)
Clerk	0.59	0.60	-0.011 (0.002)	-0.006 (0.002)
Same state	0.52	0.66	-0.141 (0.002)	-0.147 (0.002)
Foreign-born	0.11	0.10	0.008 (0.001)	0.011 (0.002)
German	0.02	0.02	0.000 (0.000)	0.002 (0.001)
Irish	0.022	0.021	0.001 (0.001)	0.002 (0.001)

Where the error counts come from

- Clerks bundled mail by destination and attached a **facing slip** naming the destination and the sending clerk.
- Railway Mail Service clerks sorted the mail in transit and recorded every misdirected item.
- Errors were double-checked by a second examiner and reported back to supervisors.
- The Railway Mail Service was an independent arm of the post office, outside the city-level rollout.

How the effect is measured

- Compare cities **before and after** the reform reaches them with cities not yet reformed.
- The rollout rule was federal and size-based, so the timing was not a local political choice.
- Errors were recorded by a body with no stake in the local office, which limits strategic reporting.
- Effects traced year by year around the arrival of the reform, and separately for each wave.

Results

Fewer delivery errors

- Reformed cities: delivery errors fall by **19 to 22 percent**.
- Holds controlling for the size rule that assigned treatment, and for a wide set of baseline city characteristics.
- No difference in trend beforehand. The gap opens at the reform, widens, then levels out.
- The same dynamic pattern in 1883 and in 1893, a decade apart.
- Not driven by any single city or state.

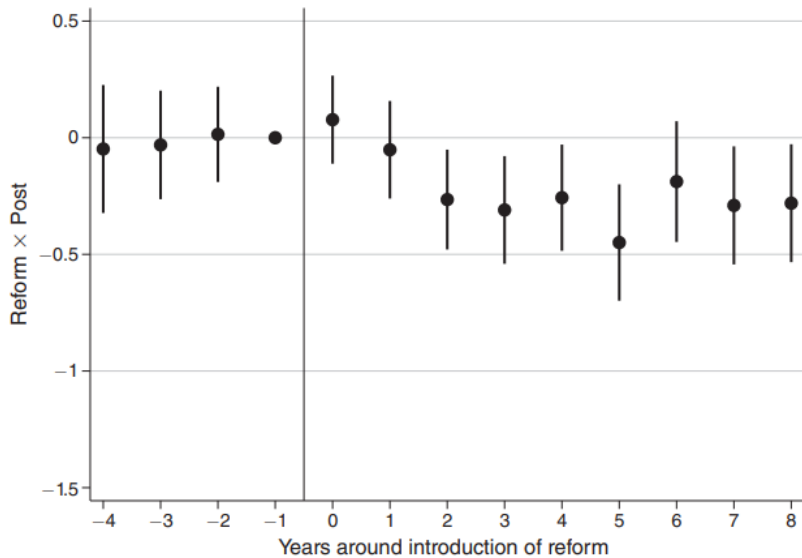
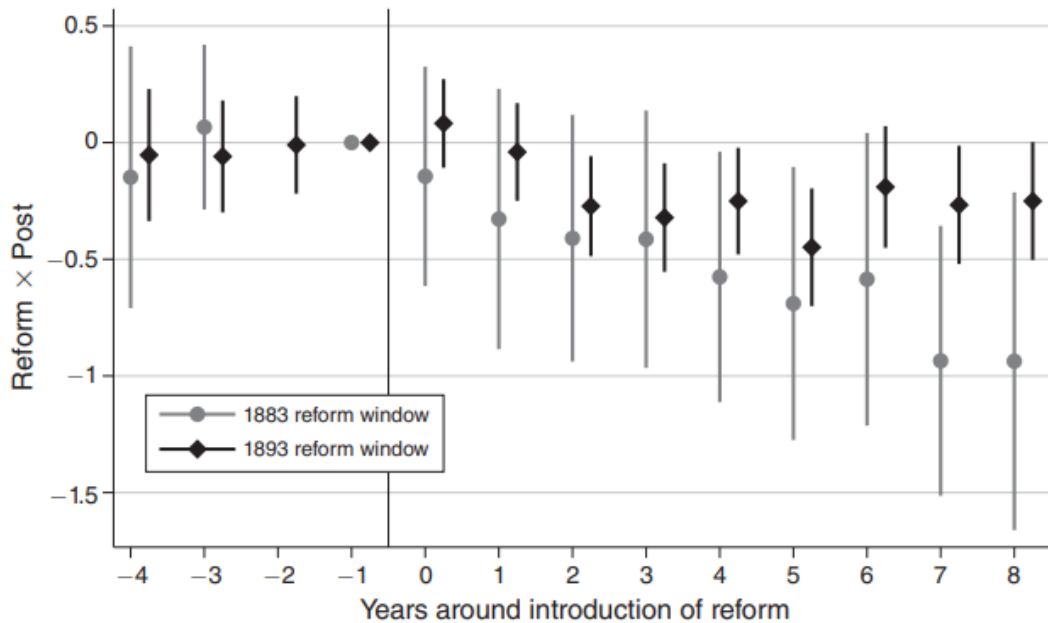


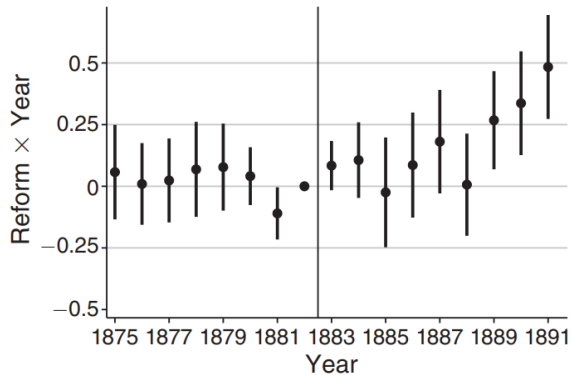
FIGURE 2. DELIVERY ERRORS FOR REFORM VERSUS NONREFORM CITIES AROUND REFORM YEARS



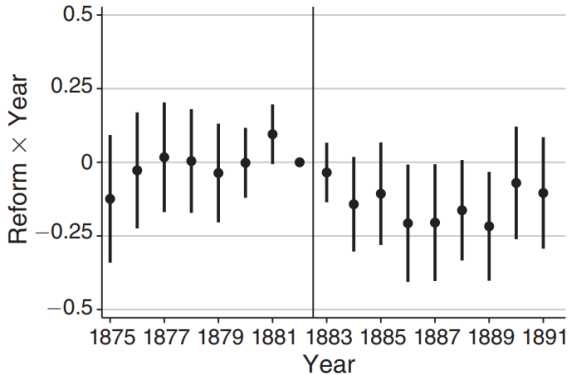
Higher productivity

- A second, independent record series: the City Free Delivery statistics, for the 1883 wave.
- Mail volume per carrier rises **8 to 14 percent**.
- Cost per volume falls 10 to 13 percent.
- Two measures from two sources, pointing the same way.

Panel A. Mail per carrier



Panel B. Cost per volume



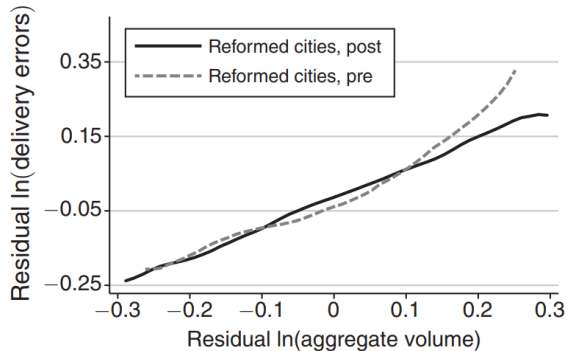
Ruling out the simple explanations

- More staff? Total postal employment moves 1.4 to 1.7 percent, indistinguishable from zero. Errors per worker fall as well.
- Different recruits? Home state, foreign birth, nationality, age, literacy, race and pay of new hires are all unchanged.
- Less work? Mail volume, if anything, rose.
- What changes is the **link between volume and errors**.

Volume and errors

- Before the reform, delivery errors rose with the volume of mail handled.
- After the reform, that relationship flattens in reformed cities.
- No comparable change among unreformed cities over the same years.
- The reform altered the “technology” of production rather than the workload.

Panel A. Reformed cities



Panel B. Unreformed cities

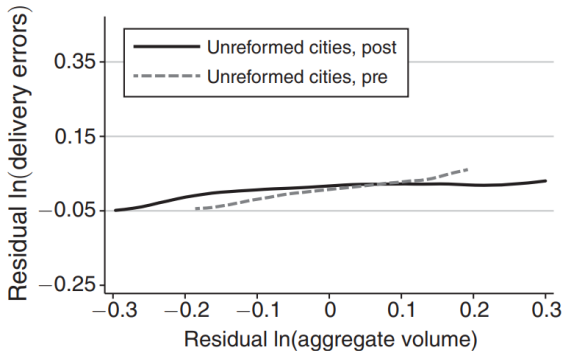
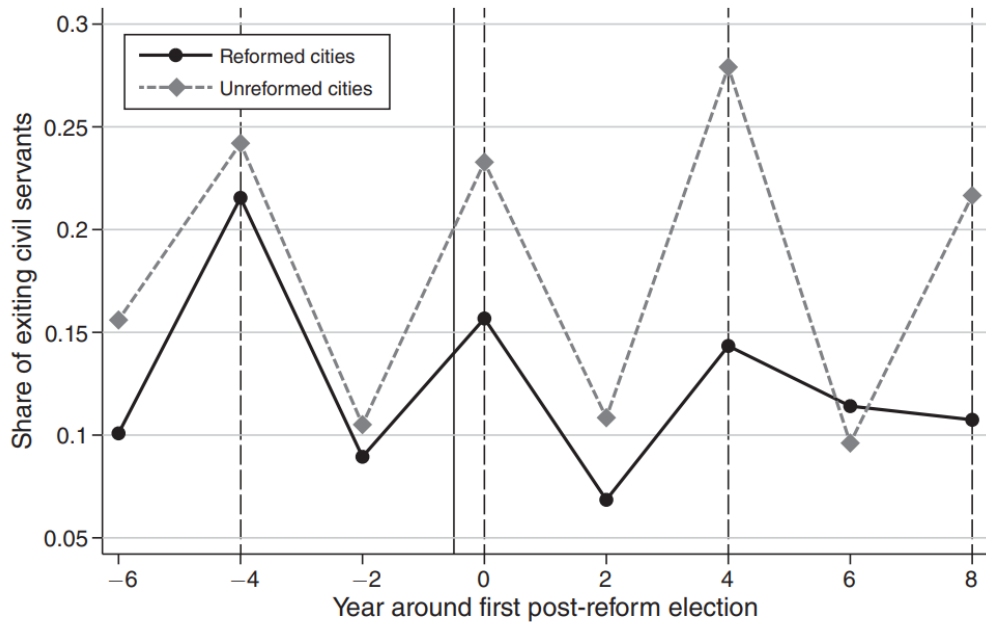


FIGURE 5. DELIVERY ERRORS AND AGGREGATE MAIL VOLUME AROUND THE REFORM

Turnover and the electoral cycle

- Serving at the pleasure of politicians meant insecure tenure. Hoogenboom (1959): “morale was low in a civil service largely composed of misfits employed on a temporary basis”.
- Annual exit rate among postal workers: **17 percent**, spiking in the years after presidential elections.
- The reform cuts the exit rate by about 4 points, a relative decline of 21 to 23 percent.
- Mean years of service rise by 0.8 to 1.0, from a base of 1.7.
- The electoral spike in exits flattens once a city is reformed.



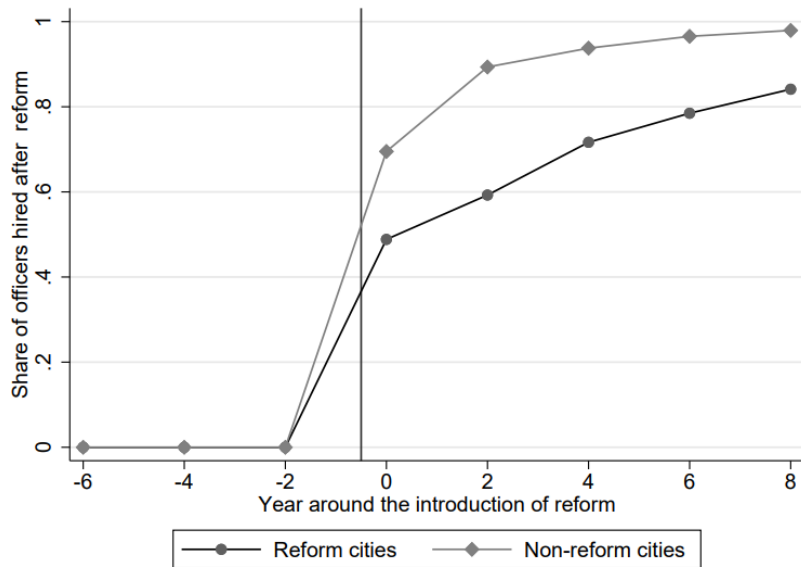
The gains land in election years

- The fall in delivery errors is driven by **election years**.
- So are the gains in volume per carrier and in cost per volume.
- Those were the years when patronage turnover previously destroyed accumulated experience.
- Retention improves for everyone serving in a reformed office, whatever their year of hire.

Selection was too slow to be the story

- Competitive exams applied only to new entrants, so replacing the existing staff was a protracted process.
- Eight years after the rules took effect, close to **20 percent** of the average reformed office were still patronage appointees.
- Lower exit is not confined to the post-reform entry cohort: it holds for those appointed under patronage too.
- The operating margin was the limit on arbitrary firing.

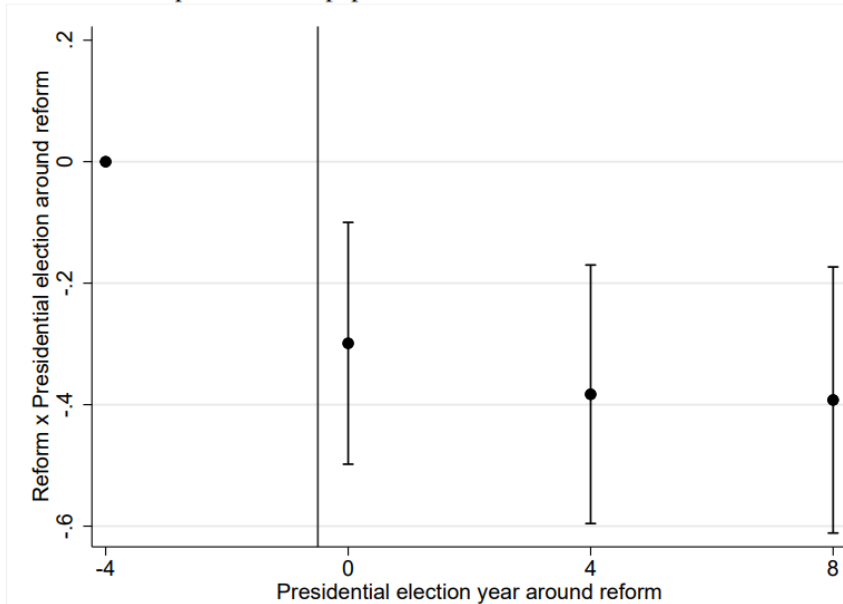
Figure A4: Share of civil servants hired after the reform event, by reform status



A side effect: weaker local parties

- Appointments were formally federal, in practice delegated to local politicians, and a source of political rents.
- Fewer patronage jobs meant less to trade for local political and financial support.
- Proxy for the strength of local parties: explicitly **partisan newspapers**, whose editors were often party officers.
- Partisan papers per city fall by 0.2 against a mean of 1.8, an 11 percent decline. Independent papers rise.
- The partisan share of circulation falls by about 4 points.

Figure A10: Number of political newspapers in reform vs. non-reform cities around reform year



In short: Aneja and Xu

- The Pendleton Act improved delivery and productivity in the largest civilian agency of the federal state.
- The channel was severing the tie between politicians and rank and file bureaucrats.
- Stable careers rather than a different pool of recruits.
- The authors call it a “double dividend”: a more effective bureaucracy, and weaker local parties.

Same reform, another agency

Moreira and Pérez (2024), *Civil Service Exams and Organizational Performance: Evidence from the Pendleton Act*, AEJ: Applied

Civil Service Exams and Organizational Performance: Evidence from the Pendleton Act
Diana Moreira and Santiago Pérez
NBER Working Paper No. 28665
April 2021, Revised January 2022
JEL No. H83,N41

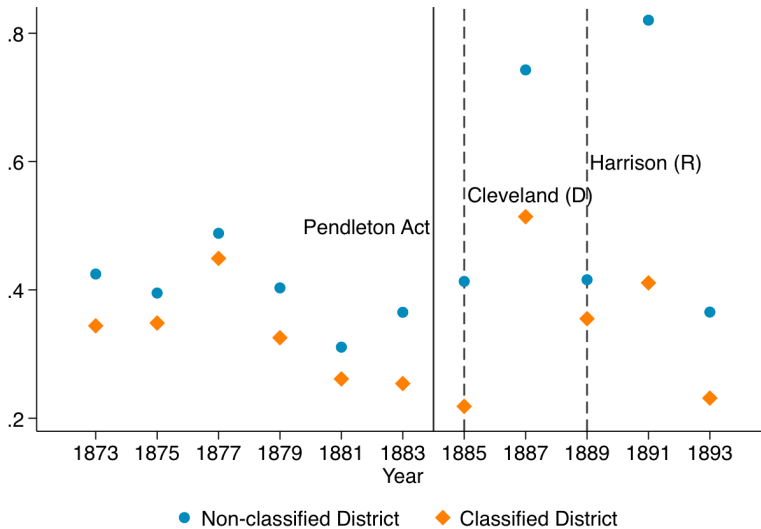
ABSTRACT

Competitive exams are a standard method for selecting civil servants. Yet, there is limited evidence on their effectiveness. We digitize personnel and financial data to study the impacts of the 1883 Pendleton Act, which mandated exams for some employees in the largest US customs-collection districts. Although the act improved targeted employees' professional background and reduced turnover, it did not increase cost-effectiveness in customs revenue collection. Moreover, it incentivized hiring in exam-exempted positions, distorting districts' hierarchical structures. These results illustrate how, by triggering countervailing organizational responses, policies that succeed at improving specific organizational aspects might nevertheless fail to improve overall

The setting: the customs houses

- Customs duties were the federal government's main source of revenue. Collecting them was the job of the Customs Service.
- The Pendleton Act reached collection districts with **at least 50 employees**: 11 districts by 1883.
- Within them, exams only for positions paying **\$900 or more**. The collectors who ran each district stayed political appointees.
- Data: personnel records from the *Official Register*, 1871 to 1893, linked to the census; receipts and expenses from the Treasury's annual reports.
- Comparison: the 11 reformed districts against the rest, before and after 1883.

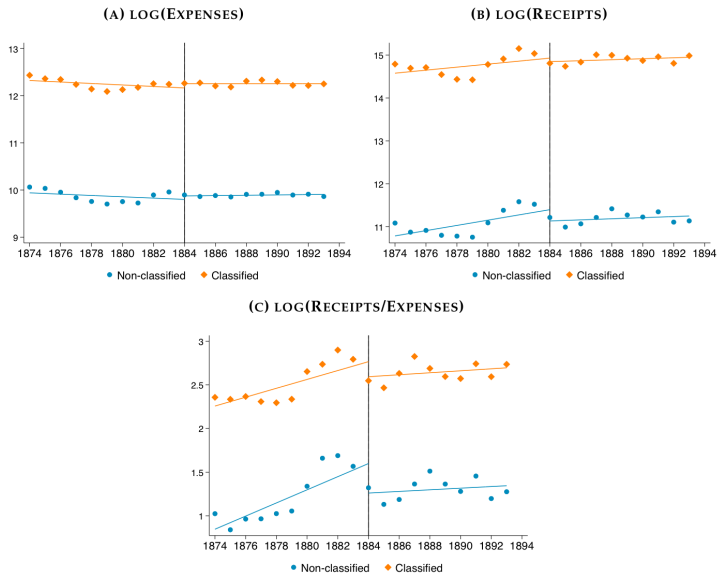
FIGURE 3: PROBABILITY OF TURNOVER



What changed: who was hired, and for how long

- New hires in exam positions were **11 percentage points** less likely to come from an unskilled job.
- The two-year turnover rate was **27 percent lower** in reformed districts.
- The drop is concentrated in the years after the White House changed party (1885, 1891): the same election-cycle pattern as in the post office.

FIGURE 5: AVERAGE EXPENSES AND RECEIPTS, 1874-1893

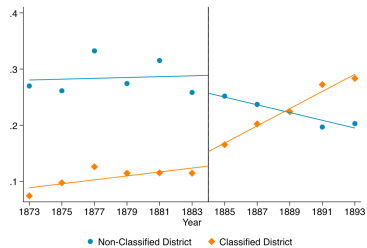


What did not change: revenue per dollar spent

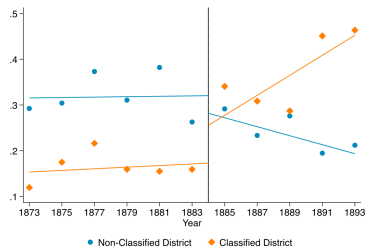
- Receipts and expenses move in parallel in reformed and unreformed districts after 1883.
- **No improvement** in revenue collected per dollar spent, even twenty years on.
- Better staff, more stable careers, same cost-effectiveness.

FIGURE 7: SHARE OF EMPLOYEES BELOW THE EXAM CUTOFF

(A) STOCK OF EMPLOYEES



(B) FLOW OF NEWLY-HIRED EMPLOYEES



Why: the holes in the reform

- Positions under \$900 stayed exempt. Reformed districts hired more of them: the share of staff below the cutoff rose from about **12 to 28 percent** between 1883 and 1893, while it fell elsewhere.
- Low-paid staff had weaker backgrounds, and the districts' hierarchy was distorted from below.
- The collectors at the top were never reformed, and they mattered: district outcomes shifted when a collector died in office and was replaced.

In short: Moreira and Pérez

- Same law, different agency, different answer.
- Exams delivered better qualified and more stable employees, but not a cheaper customs service.
- A reform that fixes one part of an organization can be undone by the parts it leaves untouched: exempt positions and political bosses.
- Post office versus customs: mail is delivered by the rank and file; revenue collection runs through the collector at the top.

The same questions today

Judges and school principals in Brazil, and hiring in firms

Selecting Top Bureaucrats: Admission Exams and Performance in Brazil*

Ricardo Dahis [†]

Laura Schiavon [‡]

Thiago Scot [§]

January 23, 2023

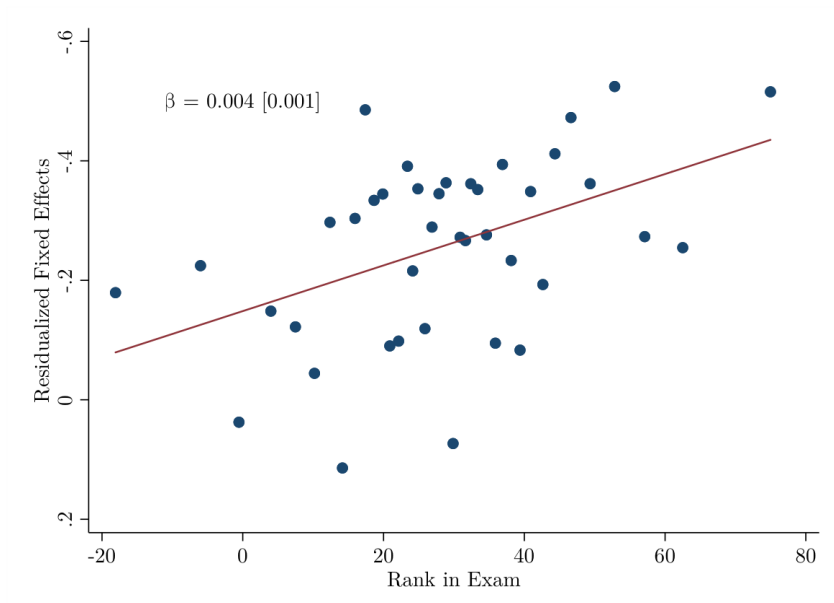
Abstract

In the absence of strong incentives, public service delivery crucially depends on bureaucrat selection. Despite wide adoption by governments, it is unclear whether civil service examinations reliably select for job performance. We investigate this question focusing on state judges in Brazil. Exploring monthly data on judicial output and cross-court movement, we estimate that judges account for at least 23% of the observed variation in number of cases disposed. With novel data on admission examinations, we show that judges with higher grades perform better than lower-ranked peers. Our results suggest competitive examinations can be an effective way to screen candidates.

Dahis, Schiavon and Scot: do exams pick good judges?

- Brazilian state judges enter through a *concurso*: multiple choice, written exam, decision writing, oral exam, titles.
- Monthly data on every state judge, 2009 to 2015: cases disposed on the merits. Judges move across courts, so judge and court can be told apart.
- Huge dispersion: a judge at the 90th percentile disposes of **eight times** as many cases as one at the 10th. The judge, not the court, explains at least **23 percent** of the variation.
- Exam grades for over 2,800 judges, 79 recruitment waves in 24 states. Within the same exam, those in the **top fifth** dispose of about **20 percent more cases**.
- The Pendleton question, answered for Brazil: the exam does screen, even among very qualified candidates.

Figure 5: Binscatter of residual ranks, conditioning on Concurso FE



Political Turnover, Bureaucratic Turnover, and the Quality of Public Services *

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Laura Trucco

Airbnb[§]

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August 12, 2018

Abstract

We study how political turnover in mayoral elections in Brazil affects public education provision. Exploiting a regression discontinuity design for close elections, we find that municipalities with a new party in office subsequently have test scores that are .05–.08 standard deviations lower. Party turnover leads to a sharp increase in the replacement rate of headmasters and teachers in schools controlled by the municipality. In contrast, turnover of the mayor's party does not impact local (non-municipal) schools. These findings suggest that political turnover can adversely affect the quality of public services when the bureaucracy is not shielded from the political process.

Akhtari, Moreira and Trucco: politics and school principals

- Municipal schools in Brazil: the mayor appoints headmasters and hires teachers.
- Compare municipalities where the incumbent party barely lost the mayoral election with those where it barely won.
- When the party changes, the share of schools with a new headmaster jumps from about **45 to 73 percent**; the share of new teachers rises too.
- Test scores (*Prova Brasil*) fall by **0.05 to 0.08 standard deviations** in the following years.
- No such effect in state schools, which the mayor does not control.
- The spoils system of the 1880s post office, observed in Brazil in the 2010s.

Figure 4: Political Turnover and School Personnel Changes

(a) Headmaster Replacement

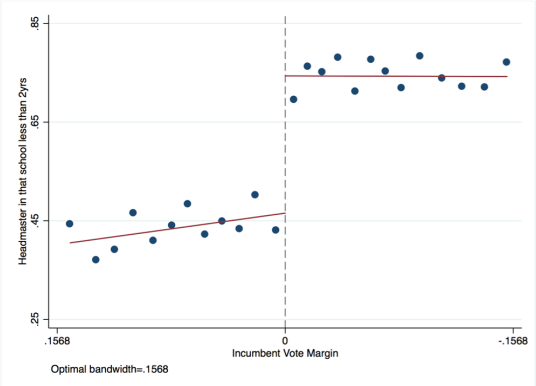
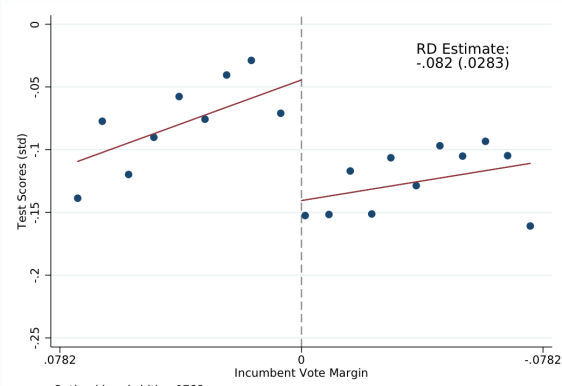


Figure 3: Political Turnover and 4th and 8th Grade Test Scores

(a) 4th Grade Test scores



Discretion in Hiring

Mitchell Hoffman, Lisa B. Kahn, and Danielle Li

NBER Working Paper No. 21709

November 2015, September 2017

JEL No. J24,M51

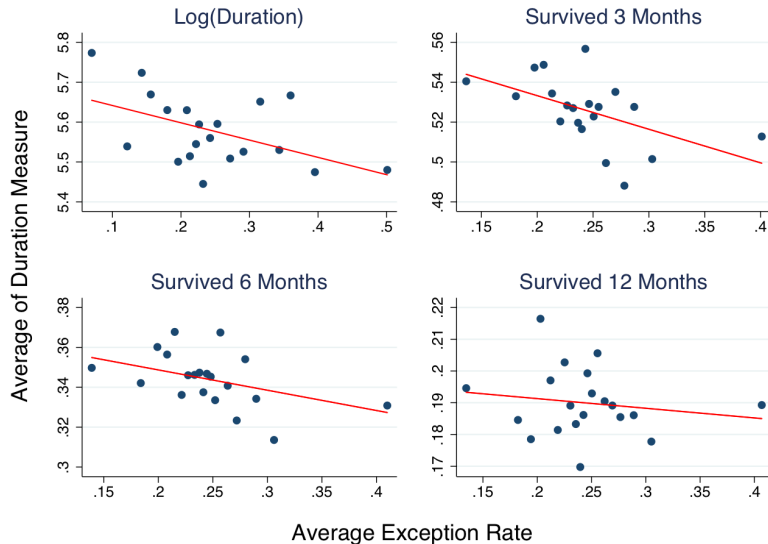
ABSTRACT

Job testing technologies enable firms to rely less on human judgement when making hiring decisions. Placing more weight on test scores may improve hiring decisions by reducing the influence of human bias or mistakes but may also lead firms to forgo the potentially valuable private information of their managers. We study the introduction of job testing across 15 firms employing low-skilled service sector workers. When faced with similar applicant pools, we find that managers who appear to hire against test recommendations end up with worse average hires. This suggests that managers often overrule test recommendations because they are biased or mistaken, not only because they have superior private information.

Hoffman, Kahn and Li: tests versus managers in firms

- 15 firms hiring low-skilled service workers, nearly **266,000 hires**. The median worker stays 99 days.
- A job test sorts applicants into green, yellow and red. Greens stay 11 percent longer than yellows, yellows 20 percent longer than reds.
- Introducing the test raises tenure of new hires by just over **25 percent**.
- Managers can overrule the test. Those who make more exceptions end up with **shorter-lived hires**: discretion added bias, not information.
- Google, 2013: tens of thousands of interview scores compared with later performance. Its head of people operations called brainteasers “a complete waste of time”: they “don’t predict anything”. Structured interviews replaced them.

FIGURE IV: MANAGER-LEVEL EXCEPTION RATES AND POST-TESTING JOB DURATIONS



Public R&D and Brazilian agriculture

Akerman, Moscona, Pellegrina and Sastry (2025), *Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution*

The question

- Standard view: innovation happens in a few frontier countries, and everyone else grows by **adopting** foreign technology.
- But frontier technology in agriculture and medicine is tailored to the environments of high-income countries.
- So public R&D in locally appropriate technology may be a way out of the technology mismatch trap.
- The case is Embrapa, the public research corporation created in Brazil in 1973.

Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution

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November 5, 2025

Abstract

Can public R&D investment in developing countries drive productivity growth? We study this question in the context of Brazil's agricultural revolution and the Empresa Brasileira de Pesquisa Agropecuária (Embrapa), a public research corporation established in 1973 to develop locally relevant science and technology. First, using researcher-level data compiled from the resumes of all agricultural scientists in Brazil, we show that Embrapa redirected research toward locally important staple crops and toward Brazil's particular ecological conditions, in part by sustaining productive research even in remote and research-scarce regions. Second, using municipality-level panel data from nine rounds of Brazil's agricultural census, we study how Embrapa affected agricultural production. We develop an identification strategy that exploits the staggered establishment of Embrapa's research laboratories and heterogeneous exposure to the benefits from new innovation in ecologically distinct parts of the country. We find that municipality-level exposure to Embrapa significantly increased agricultural yields, driven both by expanded input use and higher residual productivity. Consistent with effects being driven by locally appropriate innovation, exposure to Embrapa predicts larger take-up of Embrapa's new crop varieties and pronounced productivity gains only for the staple crops that were the focus of Embrapa's research. Third, combining our estimates with a model, we find that public R&D investment increased national agricultural productivity by 110% with a benefit-cost ratio of 17. Embrapa's decentralized structure, in which research labs were spread across many ecological zones instead of in a single hub, explains more than half of these gains.

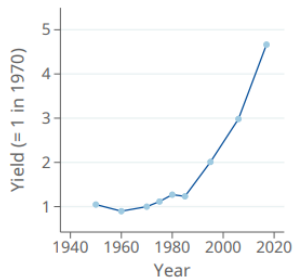
Summary

- **Question:** can public R&D in a developing country raise productivity, when frontier technology is built for other climates and soils?
- **Setting:** Embrapa, a public research corporation founded in 1973, with centers spread across Brazil's ecological regions rather than one hub. By 2010, 2,300 scientists.
- **Research:** from the resumes of every agricultural scientist in Brazil. Embrapa redirected work toward Brazilian biomes, pests and staple crops, and sustained research in remote regions.
- **Comparison:** when a center opens, municipalities ecologically similar to it against distant ones, across nine agricultural censuses.
- **Result:** more exposure, about **12 percent** higher output per unit of farm area, via more inputs, Embrapa's own varieties, and gains only in the focus crops.
- **Aggregate:** productivity up **110 percent**, 39 percent of all growth since 1970; benefit-cost ratio of **17**. A single center in Brasilia would have delivered 70 percent and a ratio of 11.
- **Lesson:** where the state put its research capacity mattered as much as how much it spent.

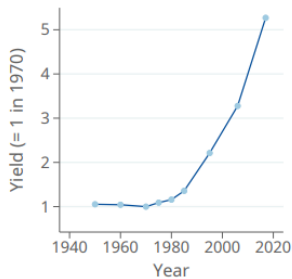
Brazilian agriculture before the 1970s

- Yields for maize, wheat and rice close to flat between 1950 and 1970.
- Already a net food importer, increasingly reliant on foreign donations and **food aid**.
- The Green Revolution did not reach Brazil: new varieties suited neither the acidic soils nor largely rainfed farming.
- Domestic research was concentrated in wealthier states and in cash crops. By 1960 more than half the non-fruit varieties from the Campinas institute were coffee, sugarcane and cotton.

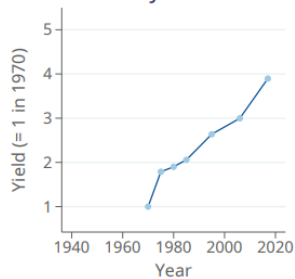
Maize



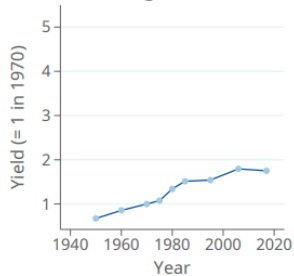
Rice



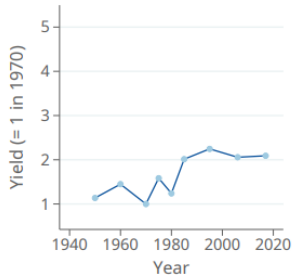
Soybeans



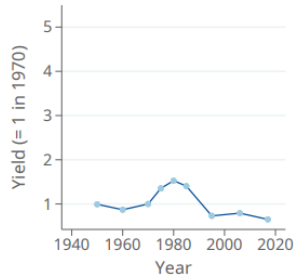
Sugarcane



Coffee



Cocoa



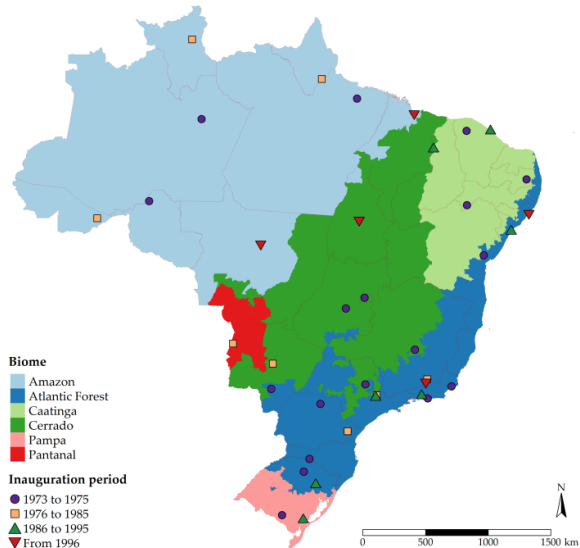
Embrapa: origins and design

- Eliseu Alves, one of its founders: “The potential existed, but there was no science capable of generating technology suited to what we needed”.
- The design group named two bottlenecks: the **geographic centralization** of research, and the lack of trained personnel and careers.
- Three organizational principles: scale, geographic scope, and the legal form of a public corporation.
- Centers opened in regions with no research tradition. Priority regions: the Northeast, the Cerrado and the Amazon.

Scale

- By the 2000s, Embrapa's spending as a share of agricultural GDP was comparable to all public agricultural R&D in the United States.
- Budget in 2010: 1.9 billion reais, about 1.15 billion dollars. The USDA research budget that year was 2.3 billion dollars.
- About **2,300 agricultural scientists** in 2010, 40 percent of all those in the country.
- Stated goal: to study every region and ecosystem of Brazil.

Figure 2: Embrapa's Research Centers



The Cerrado and soy

- The Cerrado: two million square kilometers of tropical savanna, very low soil pH, high aluminum saturation.
- Borlaug on the view in the global research community: “nobody thought these soils were ever going to be productive”.
- Embrapa Cerrados, 1975: soil chemistry and **liming**, plus bacterial nitrogen fixation. Two Embrapa researchers later received the World Food Prize for these strands.
- Embrapa Soja, also 1975, set out to tropicalize the soybean. A first variety adapted to tropical latitudes followed in 1980.
- By the 2006 census the Cerrado supplied 42 percent of national soy, against no more than 5 percent of any of these crops in the 1970s.

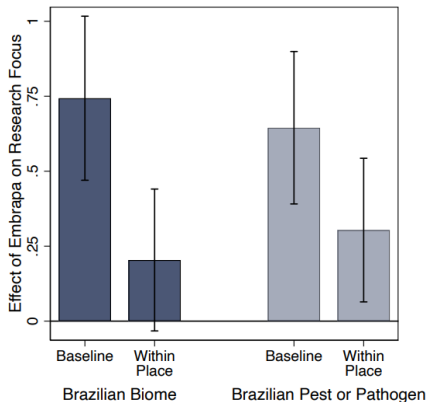
The data

- Founding year, location and annual budget of every Embrapa center, obtained by transparency request.
- The Lattes CV platform: 35,602 agricultural researchers and about 1.3 million articles, with titles coded by crop, biome and pest.
- Nine rounds of the Agricultural Census, 1960 to 2017, digitized from scans for the early years.
- Nine **agro-climatic characteristics** per municipality: temperature, precipitation, elevation, ruggedness, growing season, soil acidity and texture.

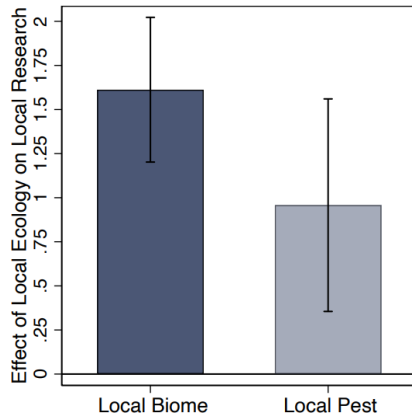
Embrapa redirected research

- Embrapa articles mention a Brazilian biome **74 percent** more often than the sample average, and a Brazilian pest or pathogen 64 percent more.
- Much of that is about where the research is done: the gap shrinks sharply once location is held fixed.
- Researchers study the biome they sit in and the pests present in their state.
- Embrapa also shifted attention to staple crops (beans, cassava, maize, rice, soy, wheat), and that shift survives holding location and the researcher fixed.

Figure 3: The Direction of Research Across Ecological Conditions

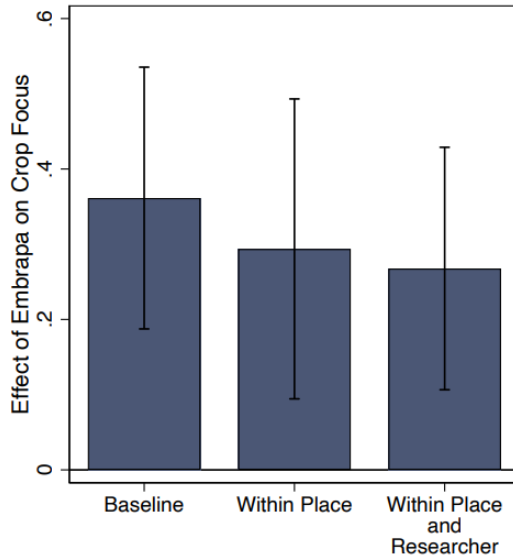


(a) Embrapa's Effect on Topic Focus ($p \times i \times t$)



(b) Local Research Focus ($i \times k$)

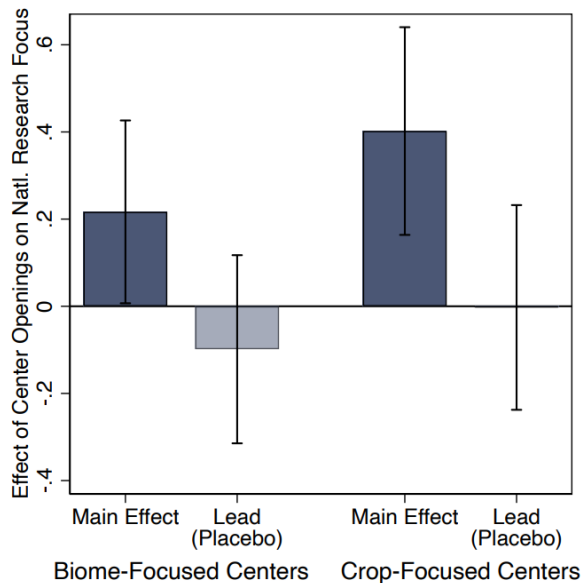
Figure 4: The Direction of Research Across Crops



No crowding out

- Embrapa might simply have absorbed research that would have happened anyway.
- Opening a center focused on a crop or a biome raised **national** research on that topic.
- Part of the increase comes from researchers not employed by Embrapa.
- No movement in the year before an opening. If anything there was crowd in.

Figure 5: Embrapa and the National Direction of Research



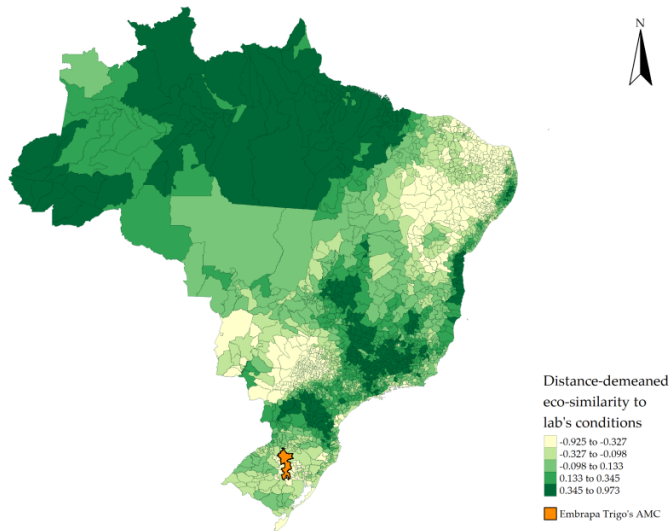
Productive research far from the hubs

- Centers went to places with little pre-existing research capacity and no agglomeration of scientists.
- Researchers move across both employers and locations, so the two effects can be separated.
- Employment at Embrapa raised research output, and by most in the **remote regions**.
- Enough to more than offset the penalty of working away from the established hubs.
- The same holds restricting to high-impact international journals.

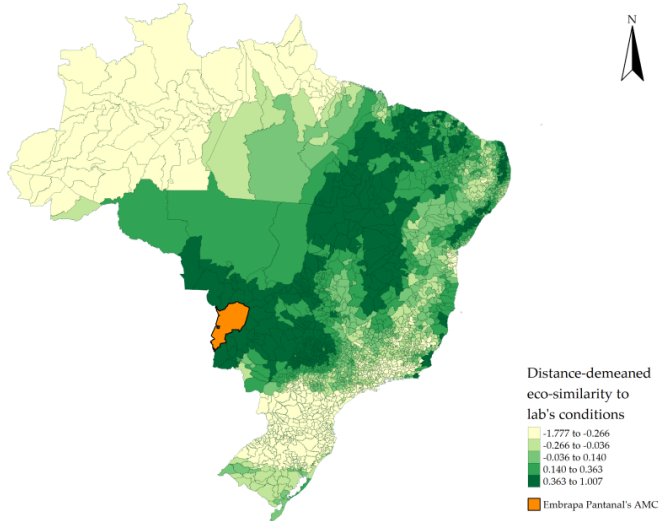
Measuring exposure to Embrapa

- Technology developed for one ecology travels to similar ecologies, not to distant ones.
- **Ecological similarity** between each municipality and each center, built from the nine agro-climatic characteristics.
- Exposure of a municipality in a given year: the highest similarity among centers open by that year.
- Exposure rises as Embrapa spreads into new ecological zones.

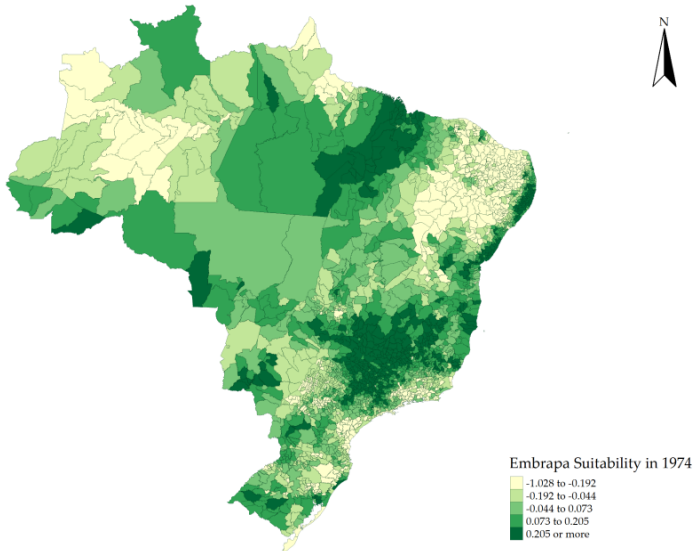
(a) Similarity to Embrapa Trigo



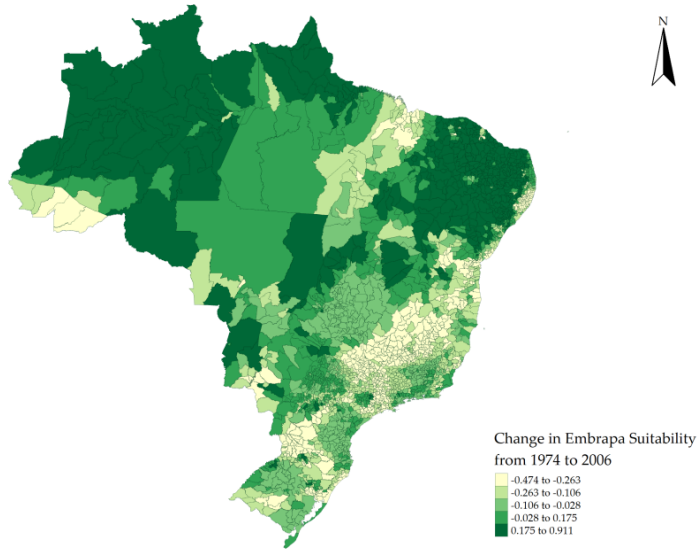
(b) Similarity to Embrapa Pantanal



(c) Embrapa Exposure in 1974



(d) Change in Exposure, 1974 to 2006



The comparison

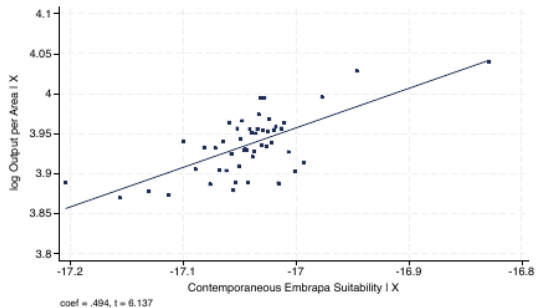
- When a center opens, compare municipalities ecologically **similar** to it with ecologically distant ones.
- Physical proximity is controlled for, and the result survives dropping municipalities within 100 km of any center.
- Assumption: similar and distant municipalities were on the same trend before the center opened.
- Placebo expansions of Embrapa across time and space put the real estimate in the right tail.

Result: higher productivity

- One standard deviation more exposure raised production value per unit of farm area by about **12 percent**.
- The same holds for land values, for crop yields, and for a broader productivity measure net of labor and inputs.
- No pre-trend over 1960 to 1970.
- The effects accumulate: the long-run effect is 30 to 40 percent larger than the within-decade effect.

Figure 7: Embrapa Affects Productivity Contemporaneously, with no Anticipation

(a) Contemporaneous Exposure



(b) Future Exposure

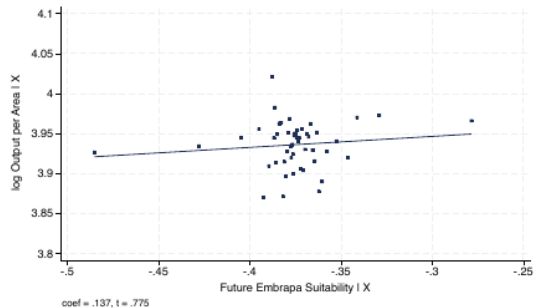
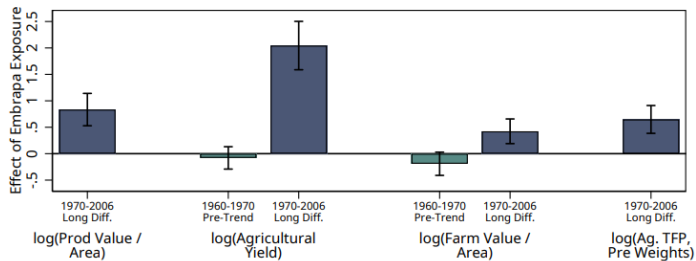


Figure 8: Long Difference Estimates and Pre-Existing Trends

(a) Unweighted



(b) Weighted by Agricultural Area

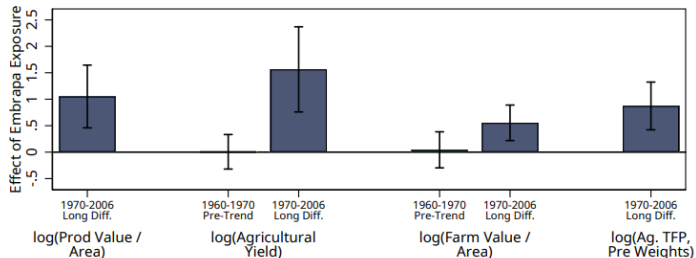
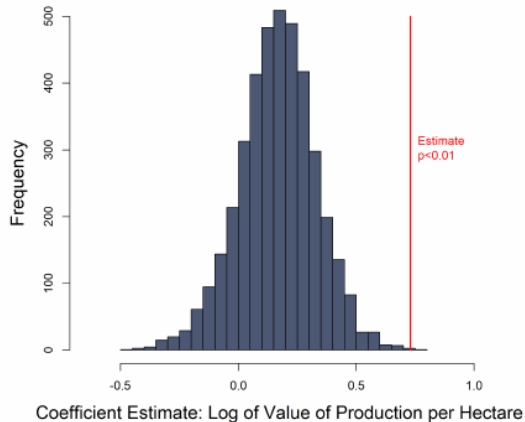
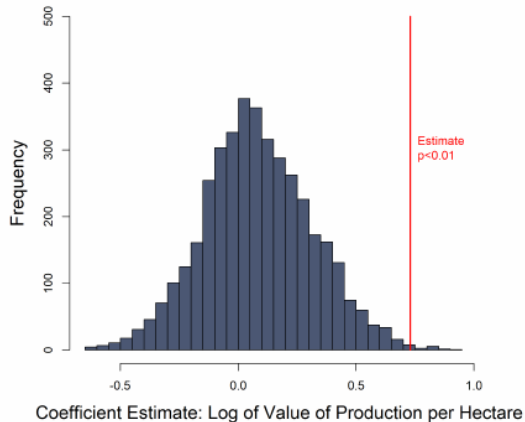


Figure 9: Falsification Tests



(a) Randomizing Across Geography and Time

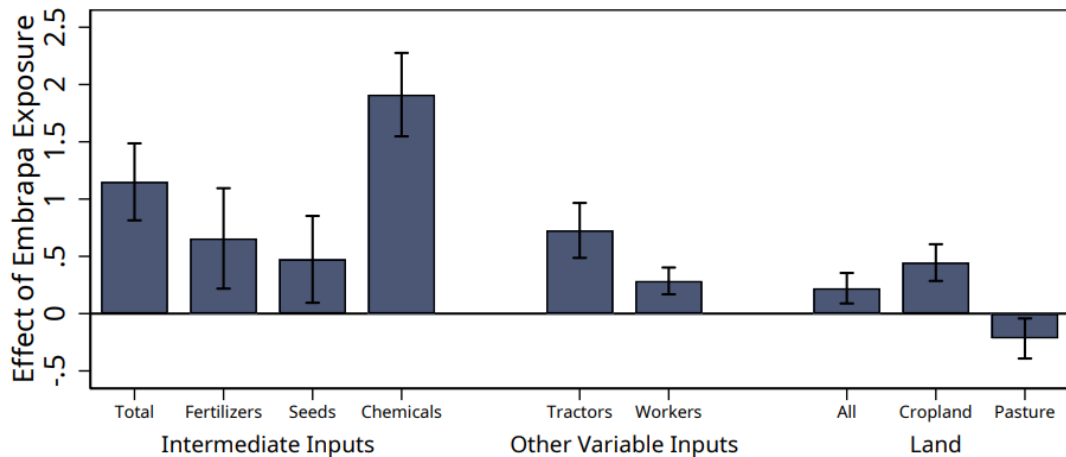


(b) Randomizing Across Time Only

Mechanisms

- Spending rises on exactly the inputs Embrapa worked on: fertilizers, seeds and chemical defenses.
- Cropland expands, over half of it converted from pasture. Labor use rises far less than input use.
- Embrapa's own **seed varieties** spread to ecologically similar municipalities even when far away: one standard deviation more similarity, about 50 percent more area planted.
- Output and yields rise for the focus crops and are flat for the rest, even within the same municipality and year.

Figure 10: Embrapa Exposure and Agricultural Inputs

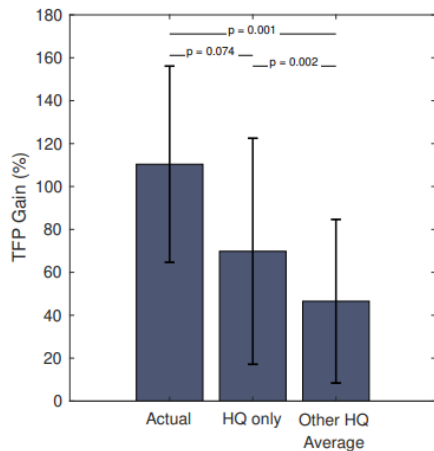


Benefits and costs

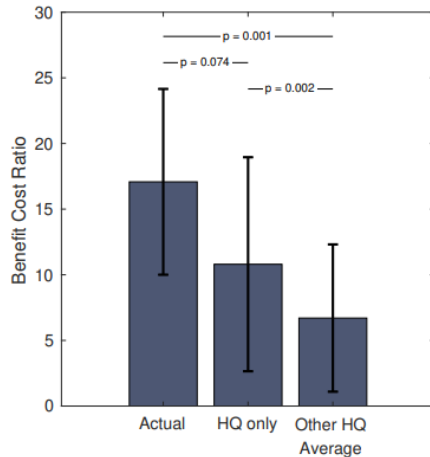
- Combining the estimates with a model: Embrapa raised aggregate agricultural productivity by **110 percent**.
- That is 39 percent of all Brazilian agricultural productivity growth since 1970.
- Benefit-cost ratio of 17, against a peak cost of about 1 percent of Brazil's agricultural GDP.
- For comparison, Jones and Summers put overall US public R&D at 5, and Griliches put hybrid corn at 7.

Figure 11: Productivity Gains and Benefit-Cost Analysis of Embrapa

(a) Productivity Gains



(b) Benefit-Cost Ratio



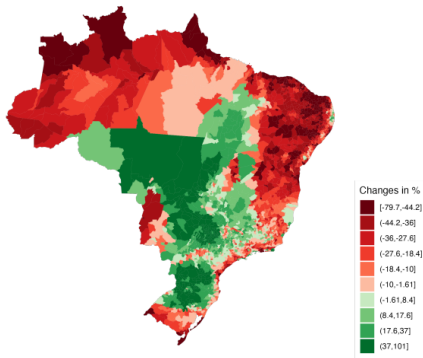
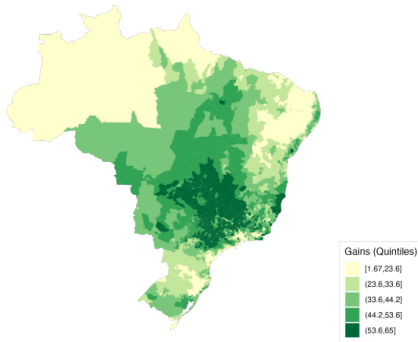
The value of spreading out

- Counterfactual: the same budget concentrated in a single center in Brasilia.
- The productivity gain falls from 110 to 70 percent, and the benefit-cost ratio from 17 to 11.
- A single center at the average alternative location achieves less than half the observed benefit.
- Brasilia is ecologically close to much of the country, so it is the best single site, which is consistent with the choice made.
- **Geographic scope** carried much of the return, over and above the level of spending.

Figure 12: Geographic Distribution of the Productivity Gains of an Embrapa Center

(a) Initial Aggregate Productivity Gain

(b) Change in Aggregate Productivity Gain



In short: Akerman and co-authors

- Public R&D aimed at local conditions raised Brazilian agricultural productivity substantially.
- The mechanism was developing and diffusing technology matched to each region's ecology.
- The return was high by the standards of the literature on public R&D.
- Where the research capacity was placed did much of the work.

Across the three papers

- Aneja and Xu: shielding rank and file bureaucrats from politics improved what the state delivered.
- Moreira and Pérez: the same reform did not pay off where the top stayed political and the exemptions were easy to exploit.
- Akerman and co-authors: where the state placed its research capacity determined which technology got made.
- All three measure the state through an **output**: mail delivered correctly, revenue per dollar spent, crop yields.

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